

Virtual Machines: A Beginner's Guide

This article is a beginner's tutorial on setting up a virtual machine and performing some basic operations on it.



in an organisational setup remain underutilised. Virtualisation helps in hardware consolidation which, in turn, improves hardware utilisation.

Easier software shipping:

Virtualisation makes software shipping effortless. Without virtualisation one has to go through the complex software installation and configuration process. Virtualisation allows vendors to ship software as pre-configured appliances.

The terminology

Each technology has its own vocabulary and virtualisation is not an exception. Let's get familiar with a few related important terms.

Host operating system (host OS):

This is the OS of the physical computer on which VirtualBox is installed.

Guest operating system (guest OS):

This is the OS that is running inside the virtual machine. VirtualBox supports a wide range of guests like Windows, Solaris, Linux, Apple, and so on

Many people wish to switch from MS Windows to Linux, but are not sure which Linux distribution is best suited to their needs. So they decide to try out a few of the most popular and user friendly Linux distributions. But wait! Isn't it a hectic and risky task to install multiple distributions on a single machine? You may lose all your data if you mess with partitions. Not necessarily. It's risky, only if you are unfamiliar with virtualisation.

Virtualisation allows the running of multiple OSs on the same machine, simultaneously. There are many virtualisation solutions currently available, such as VMware Workstation, Microsoft Hyper-V and Citrix's Xen, to name a few. However, today we will take a look at an open source virtualisation solution from Oracle called VirtualBox. It is completely free for personal and commercial use.

Benefits of virtualisation

To begin with, let us look at the benefits provided by virtualisation technology.

The capability to run multiple operating systems simultaneously: This feature greatly helps in testing software in multiple operating systems.

Better hardware utilisation: Most of the time, a number of servers



Figure 1: VirtualBox manager